



Islamic University of Technology (IUT)
Organisation of Islamic Cooperation (OIC)
Department of Electrical and Electronic Engineering



Name : Omar Faruk

ID : 200021118

Section : A2

Department : EEE

Date of Performance : 16 March 2025

Date of Submission : 16 March 2025

Course ID : EEE 4705

Course Name : Microcontroller Based System Design

Assignment No. : 01

Assignment Name : COMPLEX ENGINEERING PROBLEM

1 Introduction

The objective of this project was to design and implement a microcontroller-based oven prototype for Jurgen Oven Corporation. The system utilizes the AT89S52/AT89C51 microcontroller to enable user interaction through a keypad, countdown display on three 7-segment displays (7SDs), status indication via LEDs, dynamic messaging on an LCD, and an emergency stop feature.

2 System Design and Implementation

2.1 Hardware Configuration

- **Inputs:**
 - A 4x4 matrix keypad was configured for time entry and control (start/stop).
 - Emergency switch.
- **Outputs:**
 - Three 7SDs were multiplexed to display the countdown.
 - An LCD displayed messages (e.g., "Preheating" for ≤ 60 seconds, rotating facts for > 60 seconds).
 - LEDs indicated the oven status (1st LED = Oven ON, 2nd LED = Oven idle/stopped).
 - A buzzer activated at countdown completion (*or on Emergency Stop*).
- **Emergency Stop:**
 - A dedicated button to pause the oven.

2.2 Software Logic

- Frequency: **13.00021118 MHz**

$$\begin{aligned} StudentID &= 200021118 \\ 11 + (22 - 12) \times \left(\frac{StudentID}{10^9}\right) \\ &= 11 + (22 - 12) \times \left(\frac{200021118}{10^9}\right) \\ &= 11 + 2.00021118 = 13.00021118 \text{ MHz} \end{aligned}$$

- Timer Input with Keypad
- LCD showing time before Oven Counter Starts
- **Start Button** to Start the Oven/Counter
 - Separate Button

- Using a Button in Keypad (ON/C)
- Use LED to show the **Running Condition of Oven** (ON/C)
- Show the Countering of Counter in 7-Segment Display
- **Condition on Time:**
 - If Timer < 60s → Show MSG_1 in LCD
 - If Timer > 60s → Every 20s show different MSG_2[...]
- When Timer/Counter/Oven Ends
 - Show End MSG
 - **Turn ON Buzzer**
 - Loop over to start
 - Use ON/C on Keypad to Stop the Buzzer and Start Over
- **Emergency Button** to Pause the Oven
 - Emergency Button
 - Show Emergency MSG in LCD with ***Remaining Time***

2.3 Challenges and Solutions

- Component Interfacing: Managing concurrent output (7SDs, LCD, LEDs) required efficient PIO ports utilization and multiplexing.
- Real-Time Updates: timers ensured seamless countdown and message updates without blocking user input.
- Coding: Managing all the components with limited RAM, Registers, and Accumulator was challenging. Also, I had to make sure that the code snippets for all components do not interfere with each other.

3 Methodologies

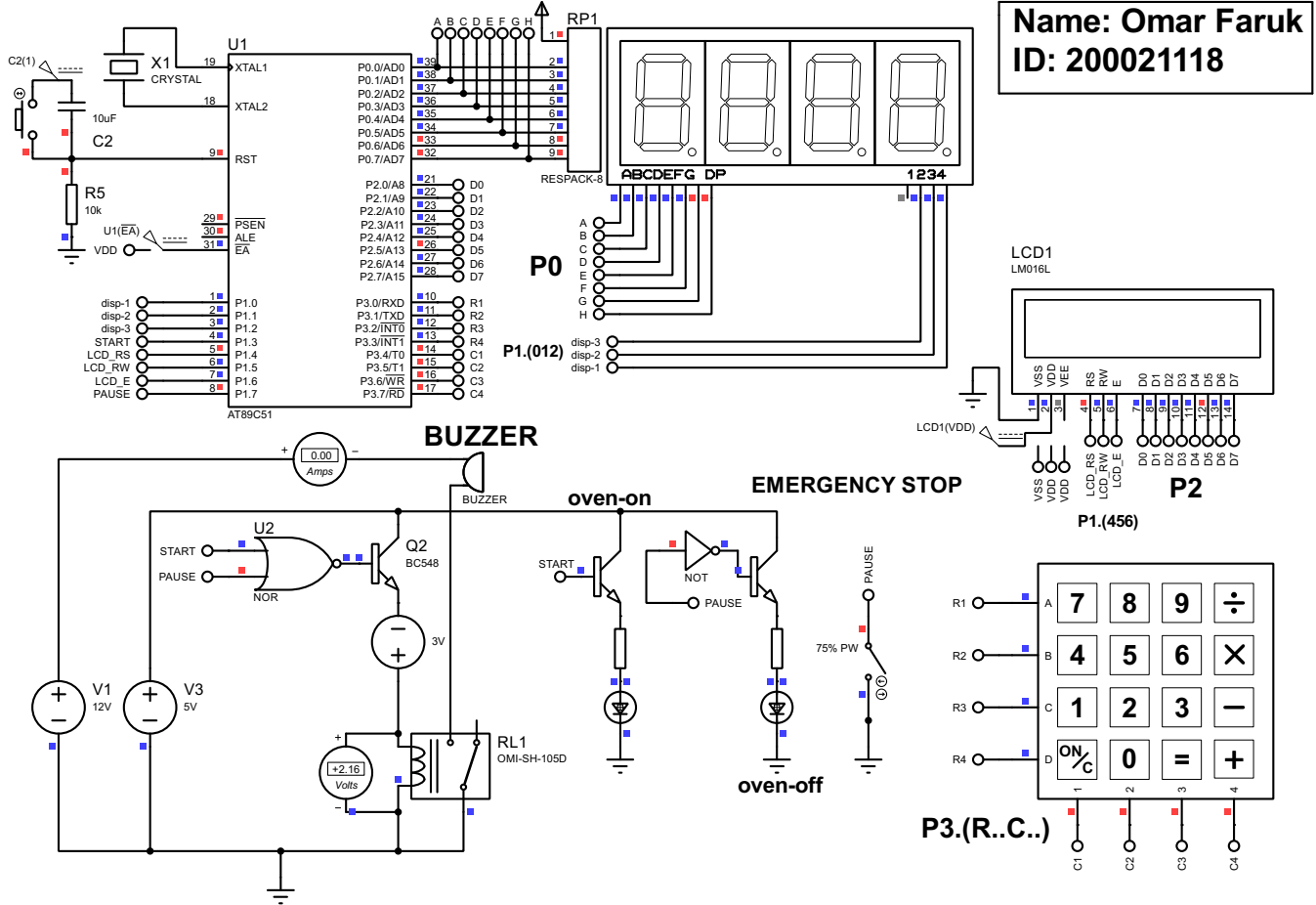


Figure 1: Proteus Circuit Diagram

Code Structure

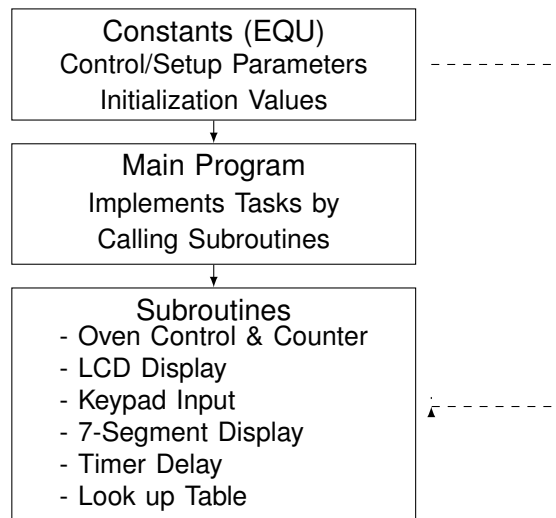


Figure 2: Assembly Code Structure

3.1 8051 Assembly Code

```
1  ORG 0
2  ; DISPLAY SETUP
3      LCD_P      EQU P2
4      LCD_RS     EQU P1.4
5      LCD_RW     EQU P1.5
6      LCD_E      EQU P1.6
7
8  ; LCD DELAY
9      LCD_DELAY_LO_ADDRESS    EQU 6FH
10     LCD_DELAY_LO_COUNT      EQU 22H
11
12 ; KEYPAD SETUP
13 ; bool ( BIT ADDRESSABLE )
14 IS_KEY_SINGLE_PRESS EQU 7FH
15 ; PORT CONNECTION ==>> C(4-1)->P(7-4) + R(4-1)->P(3-0)
16 KEYPAD_PORT          EQU P3
17
18
19 ; 7 SEG DISPLAY SETUP
20 M_7SEG_DISP_OUT_PORT EQU P0
21 M_7SEG_DISP_INDEX_0  EQU 50H
22 ; NUMBER OF DISPLAY WILL BE USED
23 M_7SEG_DISP_DIGITS_NO EQU 3
24
25 IF M_7SEG_DISP_DIGITS_NO > 0
26 M_7SEG_DISP_DIGITS_0_PIN EQU P1.0
27 ENDIF
28 IF M_7SEG_DISP_DIGITS_NO > 1
29 M_7SEG_DISP_DIGITS_1_PIN EQU P1.1
30 ENDIF
31 IF M_7SEG_DISP_DIGITS_NO > 2
32 M_7SEG_DISP_DIGITS_2_PIN EQU P1.2
33 ENDIF
34 IF M_7SEG_DISP_DIGITS_NO > 3
35 M_7SEG_DISP_DIGITS_3_PIN EQU P1.3
36 ENDIF
37
38 ; TIME_DELAY SETUP
39 TIME_DELAY_LO_COUNT EQU 55H
40 TIME_DELAY_L1_COUNT EQU 56H
41
42
43 ; OVEN_COUNTER SETUP
44 ; INPUT VARIABLE FOR DIGIT 0,1,2
45 COUNTER_INPUT_DIGIT_0 EQU 60H
46 COUNTER_INPUT_DIGIT_1 EQU 61H
47 COUNTER_INPUT_DIGIT_2 EQU 62H
48
49 ; OVEN_COUNTER LEVEL 0,1,2 FOR DIGIT 0,1,2
```

```

50     COUNTER_L0      EQU      63H
51     COUNTER_L1      EQU      64H
52     COUNTER_L2      EQU      65H
53
54     ; OVEN SETUP
55     OVEN_START_PIN   EQU      P1.3
56     OVEN_PAUSE_PIN   EQU      P1.7
57
58     ; bool ( BIT ADDRESSABLE )
59     OVEN_TIME_GREATER_60_BIT EQU 7EH
60
61
62
63     MOV SP, #70H
64     MOV PSW, #00H
65
66
67
68 ; ===== MAIN CODE START =====
69 MAIN:    ; MAIN CODE START
70         LCALL LCD_INIT      ; SETUP LCD DISPLAY
71         CLR IS_KEY_SINGLE_PRESS
72
73 GET_OVEN_READY:
74         ; SETUP OVEN PINS AS OUTPUT(0), INPUT(1)
75         LCALL OVEN_INIT
76         LCALL LCD_CLEAR
77
78         LCALL LCD_CURSOR_LINE_1
79         MOV DPTR, #OVEN_READY_MSG
80         LCALL LCD_STR_PRINT ; LCD STRING DISPLAY
81
82         LCALL LCD_CURSOR_LINE_2
83         MOV DPTR, #OVEN_TIMER_MSG
84         LCALL LCD_STR_PRINT ; LCD STRING DISPLAY
85
86 TAKE_TIMER_INPUT:
87         LCALL COUNTER_VALUE_INPUT
88
89         LCALL OVEN_CHECK_TIME_60
90
91 GET_OVEN_START:
92         ; PRESS 'ON/C' IN KEYPAD TO START/END
93         LCALL OVEN_WAIT_START_END
94
95         LCALL M_7SEG_INIT
96         LCALL OVEN_COUNTER_INIT
97
98         LCALL OVEN_COUNTER ; START COUNTER
99
100        LCALL OVEN_STOP

```

```

101         LCALL OVEN_BUZZER      ; ACTIVATES BUZZER
102
103 GET_OVEN_END:
104     LCALL LCD_CLEAR
105
106     ;LCD STRING DISPLAY
107     MOV DPTR, #OVEN_END_MSG
108     LCALL LCD_STR_PRINT
109
110     ; PRESS 'ON/C' IN KEYPAD TO START/END
111     LCALL OVEN_WAIT_START_END
112
113 GET_OVEN_REPEAT:
114     JMP GET_OVEN_READY      ; STRAT OVER
115
116
117 MAIN_END:    LJMP PROG_END;END
118 ; ===== MAIN CODE END =====
119
120
121
122 ; ===== SUBROUTINE / FUNCTIONS =====
123 ; OVEN INTEGRATE
124 OVEN_TASK_GREATER_60:
125     JNB OVEN_TIME_GREATER_60_BIT, OVEN_TASK_GREATER_60_RETURN
126     MOV A, COUNTER_L1
127     RRC A
128     JC OVEN_TASK_GREATER_60_RETURN
129     LCALL OVEN_PRINT_GREATER_60
130 OVEN_TASK_GREATER_60_RETURN:
131     RET
132
133 OVEN_COUNTER:
134     ; SECOND DELAY
135     SEC_DELAY_L0:
136         MOV R4, #0F0H
137     SEC_DELAY_L1:
138         MOV R5, #01H
139     SEC_DELAY_L2:
140         MOV R6, #01H
141     SEC_DELAY_L3:
142         ; EMERGENCY PAUSE
143         LCALL EMERGENCY_STOP
144
145         ; 7SEG DISPLAY CODE
146         LCALL COUNTER_DISPLAY_7SEG
147
148         DJNZ R6, SEC_DELAY_L3
149         DJNZ R5, SEC_DELAY_L2
150         DJNZ R4, SEC_DELAY_L1
151

```

```

152         ; CONDITION APPLIED FOR TIME > 60S
153         LCALL OVEN_TASK_GREATER_60
154
155     COUNTER_DECREMENT:
156         DJNZ COUNTER_LO, OVEN_COUNTER
157             MOV COUNTER_LO, #0AH
158         DJNZ COUNTER_L1, OVEN_COUNTER
159             MOV COUNTER_L1, #0AH
160         DJNZ COUNTER_L2, OVEN_COUNTER
161             ;MOV COUNTER_L2, #0AH
162         RET
163
164     EMERGENCY_STOP:
165         JB OVEN_PAUSE_PIN, EMERGENCY_END
166         LCALL OVEN_STOP
167         LCALL LCD_CLEAR
168         LCALL LCD_CURSOR_LINE_1
169         ;LCD STRING DISPLAY
170         MOV DPTR, #LCD_OVEN_EMERGENCY_STOP
171         LCALL LCD_STR_PRINT
172
173         LCALL LCD_CURSOR_LINE_2
174         ;LCD STRING DISPLAY
175         MOV DPTR, #LCD_OVEN_REMAINING_TIME
176         LCALL LCD_STR_PRINT
177         LCALL OVEN_LCD_TIME_PRINT
178
179         ; WAIT
180     OVEN_PAUSE: JNB OVEN_PAUSE_PIN, OVEN_PAUSE
181                 LCALL OVEN_START
182
183     EMERGENCY_END:
184         RET
185
186     OVEN_BUZZER:
187         ; NOT_START & NOT_PAUSE == OVEN_END
188         ; NOR(START , PAUSE) == OVEN_END
189         CLR OVEN_START_PIN
190         CLR OVEN_PAUSE_PIN
191         RET
192
193     OVEN_INIT:; SETUP OVEN PINS
194         CLR OVEN_START_PIN                ; OVEN START PIN AS OUTPUT(0)
195         SETB OVEN_PAUSE_PIN                ; OVEN PASUE PIN AS INPUT(1)
196         CLR OVEN_TIME_GREATER_60_BIT      ; CLEAR OVEN_TIME_LESS_60 BOOL
197         RET
198
199     OVEN_CHECK_TIME_60:
200         ; CHECK IF TIME IS GREATER THAN 60 SEC
201         MOV A, COUNTER_INPUT_DIGIT_2
202         JNZ OVEN_TIME_G_60

```



```

203     MOV A, COUNTER_INPUT_DIGIT_1
204     CLR C
205     SUBB A, #06H
206     JNC OVEN_TIME_G_60
207     CLR OVEN_TIME_GREATER_60_BIT
208     RET
209 OVEN_TIME_G_60:
210     SETB OVEN_TIME_GREATER_60_BIT
211     RET
212
213 OVEN_LCD_TIME_PRINT:
214     MOV A, COUNTER_L2
215     DEC A
216     LCALL LCD_WRITE_NUMBER
217     MOV A, COUNTER_L1
218     DEC A
219     LCALL LCD_WRITE_NUMBER
220     MOV A, COUNTER_LO
221     DEC A
222     LCALL LCD_WRITE_NUMBER
223     RET
224
225 OVEN_WAIT_START_END:
226     ; PRESS 'ON/C' IN KEYPAD TO START/END
227     ; MOV A, #'S'
228     ; LCALL LCD_WRITE_DATA
229     LCALL KEY_GET_FROM_TABLE
230     CJNE A, #'C', OVEN_WAIT_START_END
231     LCALL OVEN_START          ; START THE OVEN
232     RET
233
234 OVEN_START:
235     SETB OVEN_START_PIN      ; START THE OVEN
236     LCALL LCD_CLEAR
237     LCALL LCD_CURSOR_LINE_1
238     ;LCD STRING DISPLAY
239     MOV DPTR, #LCD_OVEN_RUNNING
240     LCALL LCD_STR_PRINT
241
242     JNB OVEN_TIME_GREATER_60_BIT, NOT_TIME_GREATER_60
243     LCALL OVEN_PRINT_GREATER_60
244     RET
245 NOT_TIME_GREATER_60:
246     LCALL LCD_CURSOR_LINE_2
247     ;LCD STRING DISPLAY
248     MOV DPTR, #LCD_NOT_TIME_G_60_STR
249     LCALL LCD_STR_PRINT
250     RET
251
252 OVEN_PRINT_GREATER_60:
253     LCALL LCD_CURSOR_LINE_2

```

```

254      ;LCD STRING DISPLAY
255      MOV DPTR, #TIME_GREATER_60_MSG
256      LCALL LCD_STR_PRINT
257
258      MOV A, COUNTER_L2
259      DEC A
260      MOV B, #0AH
261      MUL AB
262      ADD A, COUNTER_L1
263      DEC A
264      RRC A
265      LCALL LCD_WRITE_NUMBER
266      RET
267
268  OVEN_STOP :
269      CLR  OVEN_START_PIN      ; STOP  THE OVEN
270      RET
271
272  OVEN_COUNTER_INIT:
273      MOV COUNTER_L2, COUNTER_INPUT_DIGIT_2
274      MOV COUNTER_L1, COUNTER_INPUT_DIGIT_1
275      MOV COUNTER_L0, COUNTER_INPUT_DIGIT_0
276      INC COUNTER_L2
277      INC COUNTER_L1
278      INC COUNTER_L0
279      RET
280
281  COUNTER_VALUE_INPUT:
282      ; TAKE INPUT FOR INDEX 2
283      LCALL KEY_GET_NUMBER
284      MOV COUNTER_INPUT_DIGIT_2 , A
285      MOV A, COUNTER_INPUT_DIGIT_2
286      LCALL LCD_WRITE_NUMBER
287
288      ; TAKE INPUT FOR INDEX 1
289      LCALL KEY_GET_NUMBER
290      MOV COUNTER_INPUT_DIGIT_1 , A
291      MOV A, COUNTER_INPUT_DIGIT_1
292      LCALL LCD_WRITE_NUMBER
293
294      ; TAKE INPUT FOR INDEX 0
295      LCALL KEY_GET_NUMBER
296      MOV COUNTER_INPUT_DIGIT_0 , A
297      MOV A, COUNTER_INPUT_DIGIT_0
298      LCALL LCD_WRITE_NUMBER
299      RET
300
301  COUNTER_DISPLAY_7SEG:
302      MOV A, COUNTER_L2
303      DEC A
304      LCALL M_7SEG_DISP_OUT_2

```

```

305
306     MOV A, COUNTER_L1
307     DEC A
308     LCALL M_7SEG_DISP_OUT_1
309
310     MOV A, COUNTER_LO
311     DEC A
312     LCALL M_7SEG_DISP_OUT_0
313     RET
314
315 ; FOR LCD
316 LCD_STR_PRINT:
317     ;MOV DPTR, #LCD_STR
318     LCD_STR_PRINT_AGAIN:
319         CLR A                ;set A=0 (match found)
320         MOVC A, @A+DPTR ;get ASCII code from table
321         JZ LSP_END
322         LCALL LCD_WRITE_DATA
323         INC DPTR
324         SJMP LCD_STR_PRINT_AGAIN
325     LSP_END:
326         RET
327
328
329 LCD_DELAY:
330     MOV LCD_DELAY_LO_ADDRESS, #LCD_DELAY_LO_COUNT
331     LCD_DELAY_LO:
332         DJNZ LCD_DELAY_LO_ADDRESS, LCD_DELAY_LO
333         RET
334
335 LCD_CMD:
336     LCALL LCD_READY          ;send command to LCD
337     MOV LCD_P, A             ;copy reg A to port 1
338     CLR LCD_RS               ;LCD_RS=0 for command
339     CLR LCD_RW               ;R/W=0 for write
340     SETB LCD_E               ;LCD_E=1 for high pulse
341     LCALL LCD_DELAY          ;give LCD some time
342     CLR LCD_E               ;LCD_E=0 for H-to-L pulse
343     LCALL LCD_DELAY          ;give LCD some time
344     RET
345
346
347 LCD_WRITE_DATA:
348     LCALL LCD_READY          ;write data to LCD
349     MOV LCD_P, A             ;copy reg A to port1
350     SETB LCD_RS              ;LCD_RS=1 for data
351     CLR LCD_RW               ;R/W=0 for write
352     SETB LCD_E               ;LCD_E=1 for high pulse
353     LCALL LCD_DELAY          ;give LCD some time
354     CLR LCD_E               ;LCD_E=0 for H-to-L pulse
355     LCALL LCD_DELAY          ;give LCD some time

```

```

356         RET
357
358 LCD_WRITE_NUMBER:
359         LCALL LCD_READY      ;write data to LCD
360         ADD A, #30H          ;CONVERT ASCII NUM
361         MOV LCD_P, A         ;copy reg A to port1
362         SETB LCD_RS          ;LCD_RS=1 for data
363         CLR LCD_RW           ;R/W=0 for write
364         SETB LCD_E           ;LCD_E=1 for high pulse
365         LCALL LCD_DELAY      ;give LCD some time
366         CLR LCD_E            ;LCD_E=0 for H-to-L pulse
367         LCALL LCD_DELAY      ;give LCD some time
368         RET
369
370 LCD_READY:
371         SETB LCD_P.7
372         CLR LCD_RS
373         SETB LCD_RW
374
375 LCD_WAIT:
376         CLR LCD_E
377         LCALL LCD_DELAY
378         SETB LCD_E
379         JB LCD_P.7, LCD_WAIT
380         RET
381
382 LCD_INIT:
383         MOV A, #38H          ;init. LCD 2 lines, 5x7 matrix
384         ACALL LCD_CMD
385         LCALL LCD_COFF       ;display on, cursor on
386         LCALL LCD_CMD
387         RET
388
389 LCD_CURSOR_LINE_1:
390         MOV A, #80H          ; CURSOR AT BEGINNING OF LINE 1
391         LCALL LCD_CMD
392         RET
393
394 LCD_CURSOR_LINE_2:
395         MOV A, #0C0H         ; CURSOR AT BEGINNING OF LINE 2
396         LCALL LCD_CMD
397         RET
398
399 LCD_CON:
400         MOV A, #0FH          ; Display on, cursor blinking
401         LCALL LCD_CMD
402         RET
403
404 LCD_COFF:
405         MOV A, #0CH          ; Display on, cursor off
406         LCALL LCD_CMD

```

```

407     RET
408
409 LCD_CLEAR:
410     MOV A, #01H           ; Clear Display screen
411     LCALL LCD_CMD
412     RET
413
414 LCD_CD:
415     MOV A, #04H           ; Decrement cursor (shift cursor to left)
416     LCALL LCD_CMD
417     RET
418
419 LCD_CI:
420     MOV A, #06H           ; Increment cursor (shift cursor to right)
421     LCALL LCD_CMD
422     RET
423
424 LCD_CSL:
425     MOV A, #10H           ; Shift cursor position to left
426     LCALL LCD_CMD
427     RET
428
429 LCD_CSR:
430     MOV A, #14H           ; Shift cursor position to left
431     LCALL LCD_CMD
432     RET
433
434 ; FOR KEYPAD
435 KEY_GET_NUMBER:
436     ; TAKE NUMBER INPUT FROM KEYPAD
437     LCALL KEY_GET_FROM_TABLE
438     CLR C
439     SUBB A, #30H
440     RET
441
442 KEY_GET_FROM_TABLE:
443     ; TAKE ASCII VALUE INPUT FROM KEYPAD
444     MOV KEYPAD_PORT, #0F0H
445
446     LCALL KEY_CHECK_COL    ; B = COLUMN INDEX
447     ; MOV B, A
448     LCALL KEY_CHECK_ROW    ; A = ROW INDEX
449     PUSH 0F0H
450     MOV B, #04H
451     MUL AB
452     POP 0F0H
453     ADD A, B
454     MOV DPTR, #KEYPAD_DATA
455     MOVC A, @A+DPTR
456     RET
457

```

```

458 KEY_IGNORE_NOISE:
459     MOV A, KEYPAD_PORT
460     ANL A, #0FOH
461     CJNE A, #0FOH, KEY_IGNORE_NOISE_2
462     LCALL TIME_DELAY_20MS; DELAY 20MS
463 KEY_IGNORE_NOISE_2:
464     MOV A, KEYPAD_PORT
465     ANL A, #0FOH
466     CJNE A, #0FOH, KEY_IGNORE_NOISE_RETURN
467     SJMP KEY_IGNORE_NOISE
468 KEY_IGNORE_NOISE_RETURN:
469     RET
470
471 KEY_CHECK_COL:
472     MOV A, KEYPAD_PORT
473     ANL A, #0FOH
474     CJNE A, #0FOH, KEY_CHECK_SINGLE_PRESS_LABEL
475     CLR IS_KEY_SINGLE_PRESS
476     JMP KEY_CHECK_COL
477 KEY_CHECK_SINGLE_PRESS_LABEL:
478     JB IS_KEY_SINGLE_PRESS, KEY_CHECK_COL
479 KEY_CHECK_COLUMN_0:
480     MOV B, #00H
481     CJNE A, #0EOH, KEY_CHECK_COLUMN_1
482     JMP KEY_CHECK_COLUMN_END
483 KEY_CHECK_COLUMN_1:
484     INC B
485     CJNE A, #0DOH, KEY_CHECK_COLUMN_2
486     JMP KEY_CHECK_COLUMN_END
487 KEY_CHECK_COLUMN_2:
488     INC B
489     CJNE A, #0BOH, KEY_CHECK_COLUMN_3
490     JMP KEY_CHECK_COLUMN_END
491 KEY_CHECK_COLUMN_3:
492     INC B
493     ;CJNE A, #0F7H, KEY_CHECK_COLUMN_END
494     ;JMP KEY_CHECK_COLUMN_END
495 KEY_CHECK_COLUMN_END:
496     SETB IS_KEY_SINGLE_PRESS
497     RET
498
499 KEY_CHECK_ROW:
500     PUSH 0FOH
501 KEY_CHECK_ROW_0:
502     MOV B, #00H
503     SETB KEYPAD_PORT.0
504     MOV A, KEYPAD_PORT
505     ANL A, #0FOH
506     CJNE A, #0FOH, KEY_CHECK_ROW_1
507     JMP KEY_CHECK_ROW_END
508 KEY_CHECK_ROW_1:

```

```

509     INC B
510     SETB KEYPAD_PORT.1
511     MOV A, KEYPAD_PORT
512     ANL A, #0FOH
513     CJNE A, #0FOH, KEY_CHECK_ROW_2
514     JMP KEY_CHECK_ROW_END
515 KEY_CHECK_ROW_2:
516     INC B
517     SETB KEYPAD_PORT.2
518     MOV A, KEYPAD_PORT
519     ANL A, #0FOH
520     CJNE A, #0FOH, KEY_CHECK_ROW_3
521     JMP KEY_CHECK_ROW_END
522 KEY_CHECK_ROW_3:
523     INC B
524     ; SETB KEYPAD_PORT.3
525     ; MOV A, KEYPAD_PORT
526     ; ANL A, #0FOH
527     ; CJNE A, #0FOH, KEY_CHECK_ROW_END
528 KEY_CHECK_ROW_END:
529     MOV A, B
530     POP 0FOH
531     RET
532
533 ; MULTIPLE 7 SEGMENT DISPLAY
534
535 M_7SEG_INIT: ; SETUP OUTPUT(0) PORTS FOR 7 SEG DISPLAY
536     MOV M_7SEG_DISP_OUT_PORT, #00H
537
538     ; CONDITIONAL DIRECTIVES FOR MULTIPLE 7 SEG DISPLAY
539     IF M_7SEG_DISP_DIGITS_NO > 0
540         CLR M_7SEG_DISP_DIGITS_0_PIN
541     ENDIF
542     IF M_7SEG_DISP_DIGITS_NO > 1
543         CLR M_7SEG_DISP_DIGITS_1_PIN
544     ENDIF
545     IF M_7SEG_DISP_DIGITS_NO > 2
546         CLR M_7SEG_DISP_DIGITS_2_PIN
547     ENDIF
548     IF M_7SEG_DISP_DIGITS_NO > 3
549         CLR M_7SEG_DISP_DIGITS_3_PIN
550     ENDIF
551     RET
552
553 IF M_7SEG_DISP_DIGITS_NO > 0
554 M_7SEG_DISP_OUT_0:
555     LCALL GET_INDEX_VALUE
556     MOV M_7SEG_DISP_OUT_PORT, A
557     CPL M_7SEG_DISP_DIGITS_0_PIN
558     LCALL TIME_DELAY_1MS
559     CPL M_7SEG_DISP_DIGITS_0_PIN

```

```

560         RET
561     ENDIF
562     IF M_7SEG_DISP_DIGITS_NO > 1
563     M_7SEG_DISP_OUT_1:
564         LCALL GET_INDEX_VALUE
565         MOV M_7SEG_DISP_OUT_PORT, A
566         CPL M_7SEG_DISP_DIGITS_1_PIN
567         LCALL TIME_DELAY_1MS
568         CPL M_7SEG_DISP_DIGITS_1_PIN
569         RET
570     ENDIF
571     IF M_7SEG_DISP_DIGITS_NO > 2
572     M_7SEG_DISP_OUT_2:
573         LCALL GET_INDEX_VALUE
574         MOV M_7SEG_DISP_OUT_PORT, A
575         CPL M_7SEG_DISP_DIGITS_2_PIN
576         LCALL TIME_DELAY_1MS
577         CPL M_7SEG_DISP_DIGITS_2_PIN
578         RET
579     ENDIF
580     IF M_7SEG_DISP_DIGITS_NO > 3
581     M_7SEG_DISP_OUT_3:
582         LCALL GET_INDEX_VALUE
583         MOV M_7SEG_DISP_OUT_PORT, A
584         CPL M_7SEG_DISP_DIGITS_3_PIN
585         LCALL TIME_DELAY_1MS
586         CPL M_7SEG_DISP_DIGITS_3_PIN
587         RET
588     ENDIF
589
590     GET_INDEX_VALUE:
591         MOV DPTR, #M_7SEG_ARRAY
592         MOVC A, @A+DPTR
593         CPL A ; COMPLEMENT IF NEEDED
594
595
596     ; TIME CALCULATED DELAY
597     TIME_DELAY_1MS:
598         ; FOR f = 11Hz => T = 12/f = 1.09091us
599         ; ==> n = 1ms/T = 916 = 3*L0.L1 + 3*L1 + 2
600         ; ==> L1 = (916 - 2) / ( 3*L0 + 3 )
601         ; ==> L1 = 1 & L0 = 255
602         MOV TIME_DELAY_L1_COUNT, #1
603     TIME_DELAY_1MS_L1:
604         MOV TIME_DELAY_L0_COUNT, #255
605     TIME_DELAY_1MS_L0:
606         DJNZ TIME_DELAY_L0_COUNT, TIME_DELAY_1MS_L0
607         DJNZ TIME_DELAY_L1_COUNT, TIME_DELAY_1MS_L1
608         RET
609
610     TIME_DELAY_20MS:

```



```

611         ; FOR f = 11Hz => T = 1.09091us
612         ; ==> n = 20ms/T = 18333 = 3*L0.L1 + 3*L1 + 2
613         ; ==> L1 = 23 & L0 = 255
614         MOV TIME_DELAY_L1_COUNT, #23
615     TIME_DELAY_20MS_L1:
616         MOV TIME_DELAY_L0_COUNT, #255
617     TIME_DELAY_20MS_L0:
618         DJNZ TIME_DELAY_L0_COUNT, TIME_DELAY_20MS_L0
619         DJNZ TIME_DELAY_L1_COUNT, TIME_DELAY_20MS_L1
620     RET
621
622 ; ===== DATA LOOK UP TABLE =====
623     ; END OF THE STRING HAS TO BE 0
624     OVEN_READY_MSG:
625         DB 'O', 'V', 'E', 'N', ' ', 'I', 'S', ' ', 'R', 'E', 'A', 'D', 'Y',
626         ' ', 0
627         ;NOP
628     OVEN_TIMER_MSG:
629         DB 'T', 'I', 'M', 'E', 'R', ' ', ':', ' ', 0
630         ;NOP
631     KEYPAD_DATA:
632         DB '7', '8', '9', '/', '4', '5', '6', '*', '1', '2', '3', '-', 'C',
633         '0', '=', '+', 0
634         ;NOP
635     M_7SEG_ARRAY:
636         DB 3FH, 06H, 5BH, 4FH, 66H, 6DH, 7DH, 07H, 7FH, 6FH, 77H, 7CH, 39H, 5
637         EH, 79H, 71H, 0
638         ; 0 , 1 , 2 , 3 , 4 , 5 , 6 , 7 , 8 , 9 , A , b , C ,
639         d , E , F ;
640         ;NOP
641     OVEN_END_MSG:
642         DB 'O', 'V', 'E', 'N', ' ', 'T', 'A', 'S', 'K', ' ', 'D', 'O', 'N', 'E',
643         ' ', '!', 0
644     LCD_OVEN_EMERGENCY_STOP:
645         DB 'E', 'M', 'E', 'R', 'G', 'E', 'N', 'C', 'Y', ' ', 'S', 'T', 'O', 'P',
646         ' ', 0
647     LCD_OVEN_REMAINING_TIME:
648         DB 'R', 'E', 'M', 'A', 'I', 'N', 'I', 'N', 'G', ' ', ':', ' ', 0
649     LCD_OVEN_RUNNING:
650         DB 'O', 'V', 'E', 'N', ' ', 'R', 'U', 'N', 'N', 'I', 'N', 'G', 0
651     LCD_NOT_TIME_G_60_STR:
652         DB 'T', 'I', 'M', 'E', ' ', 'U', 'N', 'D', 'E', 'R', ' ', '6', '0', ' ',
653         's', 0
654     TIME_GREATER_60_MSG:
655         DB 'T', 'I', 'M', 'E', ' ', '=', '=', ' ', '2', '0', ' ', 's', ' ', 'x', ' ',
656         ' ', 0
657
658 PROG_END:
659     END

```

Listing 1: OVEN 8051 Assembly Code

4 Discussion

In the integration of hardware design, embedded programming with AT89S52/AT89C51, and real-time system management, this project successfully solved the complex demands of the oven prototype. The solution showed synchronization with knowledge traits, maintained safety regulations, and innovative engineering technologies. For increased functionality, future improvements might incorporate IoT integration and temperature control.